29 CFR 1910.1200 (OSHA HazCom 2012)

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

1.1 Product identifier

Trade name : mirage 146a™ biofunctional

Description : biofunctional
™ Trademark, Ashland or its subsidiaries, registered in various countries

Relevant identified uses of the substance or mixture and uses advised against

Recommended use : biofunctional

Details of the supplier of the safety data sheet Ashland P.O. Box 2219 Columbus, OH 43216 United States of America (USA) 1-614-790-3333 (USA) EHSProductSafety@ashland.com	**Emergency telephone** 1-800-ASHLAND (1-800-274-5263) **Regulatory information** 1-614-790-3333 (customer service) **Product Information** 1-614-790-3333 (USA)

SECTION 2. HAZARDS IDENTIFICATION

GHS Classification

Combustible Dust :

GHS label elements

Signal Word : Warning

Hazard Statements : May form combustible dust concentrations in air.

Other hazards

None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Chemical nature : organic

Hazardous ingredients

Chemical name	CAS-No.	Classification	Concentration (%)
GLYCERINE	56-81-5	Not a hazardous substance or mixture.	>=30.00 - < 50.00
2-PHENOXYETHANOL	122-99-6	Acute Tox. 4; H302 Eye Dam. 1; H318 STOT SE 3; H335	>=1.00 - < 5.00

Actual concentration is withheld as a trade secret

SECTION 4. FIRST AID MEASURES

General advice : Do not leave the victim unattended.

If inhaled : If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a physician.

In case of skin contact : Wash off with soap and plenty of water.

In case of eye contact : Remove contact lenses.
Protect unharmed eye.
If eye irritation persists, consult a specialist.

If swallowed : Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.

Notes to physician : Treat symptomatically.

SECTION 5. FIRE-FIGHTING MEASURES

Suitable extinguishing media : Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

Specific hazards during fire fighting	:	No information available.
Hazardous combustion products	:	Carbon dioxide (CO2) Carbon monoxide Nitrogen oxides (NOx) Aldehydes
Specific extinguishing methods	:	Standard procedure for chemical fires.
Further information	:	Standard procedure for chemical fires. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Special protective equipment for fire-fighters	:	In the event of fire, wear self-contained breathing apparatus.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Use personal protective equipment. Avoid dust formation.
Environmental precautions	:	Do not allow uncontrolled discharge of product into the environment.
Methods and materials for containment and cleaning up	:	Pick up and arrange disposal without creating dust. Sweep up and shovel. Keep in suitable, closed containers for disposal.

SECTION 7. HANDLING AND STORAGE

Advice on protection against fire and explosion	:	Provide appropriate exhaust ventilation at places where dust is formed.
Advice on safe handling	:	Avoid dust formation. Avoid inhalation, ingestion and contact with skin and eyes.

For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.

Conditions for safe storage : Electrical installations / working materials must comply with the technological safety standards.

Further information on storage conditions : Keep container tightly closed in a dry and well-ventilated place.
Store at room temperature in the original container.
Keep in a dry, cool place.
Keep away from direct sunlight.

Materials to avoid : No materials to be especially mentioned.

Further information on storage stability : No decomposition if stored and applied as directed.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
GLYCERINE	56-81-5	TWA (mist, respirable fraction)	5 mg/m3	OSHA Z-1
		TWA (mist, total dust)	15 mg/m3	OSHA Z-1
		TWA (Mist - total dust)	10 mg/m3	OSHA P0
		TWA (Mist - respirable fraction)	5 mg/m3	OSHA P0

Engineering measures : Provide sufficient mechanical (general and/or local exhaust) ventilation to maintain exposure below exposure guidelines (if applicable) or below levels that cause known, suspected or apparent adverse effects.

Personal protective equipment

Respiratory protection : In the case of dust or aerosol formation use respirator with an approved filter within the capabilities of the respirator/filter combination.

Where concentrations are above recommended limits or are unknown, or a cartridge type respirator is not adequate, wear a positive-pressure supplied-air respirator.

Hand protection

Material	:	butyl-rubber
Break through time	:	480 min
Glove thickness	:	> 0.5 mm
Remarks	:	The exact break through time can be obtained from the protective glove producer and this has to be observed. Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Eye protection	:	Safety glasses
Skin and body protection	:	Work uniform or laboratory coat.
Hygiene measures	:	General industrial hygiene practice.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	:	Crystalline product
Physical state	:	solid
Color	:	amber
Odor	:	characteristic
Odor Threshold	:	No data available
pH	:	5.5 - 6.5
Melting point/freezing point	:	not determined
Boiling point/boiling range	:	not determined
Flash point	:	175 °C
Evaporation rate	:	Not applicable
Flammability (solid, gas)	:	not determined

Flammability (liquids)	:	not determined
Flammability (liquids)	:	
Upper explosion limit	:	not determined
Lower explosion limit	:	not determined
Vapor pressure	:	Not applicable
Relative vapor density	:	Not applicable
Relative density	:	No data available
Density	:	not determined
Solubility(ies)		
Water solubility	:	not determined
Solubility in other solvents	:	No data available
Partition coefficient: n-octanol/water	:	not determined
Thermal decomposition	:	No data available
Viscosity		
Viscosity, dynamic	:	Not applicable
Viscosity, kinematic	:	Not applicable
Oxidizing properties	:	Not applicable
Molecular weight	:	No data available

SECTION 10. STABILITY AND REACTIVITY

Reactivity	:	No dangerous reaction known under conditions of normal use.
Chemical stability	:	Stable under normal conditions.
Possibility of hazardous reactions	:	Dust may form explosive mixture in air.
Conditions to avoid	:	Protect from frost, heat and sunlight.

Incompatible materials	:	Oxidizing agents
Hazardous decomposition products		Carbon monoxide Carbon dioxide (CO_2)

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure	:	Inhalation Eye contact Skin contact Ingestion

Acute toxicity
Not classified due to lack of data.
Product:

Acute oral toxicity	:	Acute toxicity estimate : > 5,000 mg/kg Method: Calculation method

Components:
GLYCERINE:

Acute oral toxicity	:	LD50 (Rat): 12,600 mg/kg Method: OECD Test Guideline 401
Acute inhalation toxicity	:	LC50 (Rat): > 570 mg/m3 Exposure time: 1 h Test atmosphere: vapor Assessment: Not classified as acutely toxic by inhalation under GHS.
Acute dermal toxicity	:	LD50 (Rabbit): > 10 g/kg

2-PHENOXYETHANOL:

Acute oral toxicity	:	LD50 (Rat, male): 1,394 mg/kg
Acute inhalation toxicity	:	Assessment: No adverse effect has been observed in acute inhalation toxicity tests.
Acute dermal toxicity	:	LD50 (Rat): 14,391 mg/kg

Skin corrosion/irritation
Based on available data, the classification criteria are not met.
Product:
Species: reconstructed human epidermis (RhE)

Result: No skin irritation

Species: Human
Exposure time: 48 h
Method: human patch test
Result: No skin irritation

Species: Human
Exposure time: 24 h
Method: Human repeat insult patch test (HRIPT)
Result: No skin irritation

Components:
GLYCERINE**:**
Result: No skin irritation

2-PHENOXYETHANOL**:**
Species: Rabbit
Result: No skin irritation

Serious eye damage/eye irritation
Based on available data, the classification criteria are not met.
Product:
Result: No eye irritation

Result: No eye irritation
Method: EpiOcular

Result: No eye irritation
Method: Chorioallantoic membrane vascularization assay

Result: No eye irritation
Method: Reconstructed Human Corneal Model

Components:
GLYCERINE**:**
Result: Slight, transient irritation

2-PHENOXYETHANOL**:**
Species: Rabbit

Result: Corrosive

Respiratory or skin sensitization
Skin sensitization: Based on available data, the classification criteria are not met.
Respiratory sensitization: Not classified due to lack of data.
Product:
Test Type: Human Repeat Insult Patch Testing (HRIPT)
Routes of exposure: Dermal
Species: Humans
Method: Human repeat insult patch test (HRIPT)
Result: Not a skin sensitizer.

Components:
2-PHENOXYETHANOL**:**
Species: Guinea pig
Assessment: Does not cause skin sensitization.
Method: OECD Test Guideline 406

Germ cell mutagenicity
Not classified due to lack of data.
Product:

Genotoxicity in vitro	:	Test Type: Ames test Test species: Salmonella typhimurium Metabolic activation: with and without metabolic activation Result: negative
	:	Test Type: Ames test Test species: Escherichia coli Metabolic activation: with and without metabolic activation Result: negative

Components:
2-PHENOXYETHANOL**:**

Genotoxicity in vitro	:	Test Type: Ames test Test species: Salmonella typhimurium Metabolic activation: with and without metabolic activation Result: negative

Carcinogenicity
Not classified due to lack of data.
Reproductive toxicity
Not classified due to lack of data.
Components:
2-PHENOXYETHANOL**:**

Effects on fetal development	:	Test Type: Pre-natal Species: Rat Application Route: Oral

General Toxicity Maternal: NOAEL (No observed adverse effect level): ca. 300 mg/kg bw/day
Method: OPPTS 870.3700

STOT-single exposure
Not classified due to lack of data.
Components:
2-PHENOXYETHANOL**:**
Assessment: May cause respiratory irritation.

STOT-repeated exposure
Not classified due to lack of data.
Repeated dose toxicity
Components:
2-PHENOXYETHANOL**:**
Species: Rat, male and female
NOAEL: 369 mg/kg
Application Route: Oral
Method: OECD Test Guideline 408

Species: Rabbit, male and female
NOAEL: 500 mg/kg
Application Route: Dermal

Aspiration toxicity
Not classified due to lack of data.
Further information
Product:
Remarks: No data available

Carcinogenicity:

IARC	No ingredient of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC.
OSHA	No component of this product present at levels greater than or equal to 0.1% is on OSHA's list of regulated carcinogens.
NTP	No ingredient of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP.

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity
Product:

Ecotoxicology Assessment

Short-term (acute) aquatic hazard : Not classified based on available information.

Long-term (chronic) aquatic hazard : Not classified based on available information.

Components:

GLYCERINE:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 54,000 mg/l
Exposure time: 96 h
Test Type: static test

Toxicity to daphnia and other aquatic invertebrates : LC50 (Daphnia magna (Water flea)): 1,995 mg/l
Exposure time: 48 h

2-PHENOXYETHANOL:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): 337 - 352 mg/l
Exposure time: 96 h
Test Type: flow-through test

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 500 mg/l
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202

Toxicity to algae : NOEC (Desmodesmus subspicatus (green algae)): > 500 mg/l
End point: Growth inhibition
Exposure time: 72 h
Test Type: static test

Toxicity to fish (Chronic toxicity) : NOEC (Pimephales promelas (fathead minnow)): 23 mg/l
Exposure time: 34 d
Test Type: flow-through test
Method: OECD Test Guideline 210

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Daphnia): 9.43 mg/l
Exposure time: 21 d
End point: Reproduction Test
Test Type: semi-static test
Method: OECD Test Guideline 211

Persistence and degradability

Components:

GLYCERINE:

Biodegradability : Result: Readily biodegradable.

Biodegradation: 94 %

2-PHENOXYETHANOL:
Biodegradability : Result: Readily biodegradable.
Biodegradation: 99 %
Exposure time: 28 d
Method: OECD Test Guideline 301F

No data available

Bioaccumulative potential

Components:

GLYCERINE:
Partition coefficient: n-octanol/water : log Pow: -1.76

2-PHENOXYETHANOL:
Partition coefficient: n-octanol/water : log Pow: 1.16

No data available

Mobility in soil

Components:

No data available

Other adverse effects

Product:

Additional ecological information : No data available

Components:

2-PHENOXYETHANOL:
Results of PBT and vPvB assessment : Substance is not persistent, bioaccumulative, and toxic (PBT).

: Substance is not very persistent and very bioaccumulative (vPvB).

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods

Contaminated packaging : Empty containers should be taken to an approved waste handling site for recycling or disposal.

SECTION 14. TRANSPORT INFORMATION

International transport regulations

REGULATION

ID NUMBER	PROPER SHIPPING NAME	*HAZARD CLASS	SUBSIDIARY HAZARDS	PACKING GROUP	MARINE POLLUTANT / LTD. QTY.

U.S. DOT - ROAD

Not dangerous goods

CFR_RAIL_C

Not dangerous goods

U.S. DOT - INLAND WATERWAYS

Not dangerous goods

TDG_ROAD_C

Not dangerous goods

TDG_RAIL_C

Not dangerous goods

TDG_INWT_C

Not dangerous goods

INTERNATIONAL MARITIME DANGEROUS GOODS

Not dangerous goods

INTERNATIONAL AIR TRANSPORT ASSOCIATION - CARGO

Not dangerous goods

INTERNATIONAL AIR TRANSPORT ASSOCIATION - PASSENGER

Not dangerous goods

MX_DG

Not dangerous goods

***ORM = ORM-D, CBL = COMBUSTIBLE LIQUID**

Marine pollutant		no

Dangerous goods descriptions (if indicated above) may not reflect quantity, end-use or region-specific exceptions that can be applied. Consult shipping documents for descriptions that are specific to the shipment.

SECTION 15. REGULATORY INFORMATION

TSCA list
No substances are subject to a Significant New Use Rule.

No substances are subject to TSCA 12(b) export notification requirements.

EPCRA - Emergency Planning and Community Right-to-Know

CERCLA Reportable Quantity
This material does not contain any components with a CERCLA RQ.

SARA 304 Extremely Hazardous Substances Reportable Quantity
This material does not contain any components with a section 304 EHS RQ.

SARA 311/312 Hazards : Combustible Dust

SARA 302 : This material does not contain any components with a section 302 EHS TPQ.

SARA 313 The following components are subject to reporting levels established by SARA Title III, Section 313:

2-PHENOXYETHANOL	122-99-6	1.49 %

Pennsylvania Right To Know

WATER	7732-18-5
GLYCERINE	56-81-5

New Jersey Right To Know

WATER	7732-18-5
GLYCERINE	56-81-5
2-PHENOXYETHANOL	122-99-6
BAOBAB EXTRACT	91745-12-9

California Prop. 65

This product does not contain any chemicals known to the State of California to cause cancer, birth, or any other reproductive defects.

The ingredients of this product are reported in the following inventories:

TCSI	:	On the inventory, or in compliance with the inventory
AIIC	:	Not in compliance with the inventory
DSL	:	This product contains one or several components that are not on the Canadian DSL and have annual quantity limits.
ISHL	:	Not in compliance with the inventory
ENCS	:	Personal care
KECI	:	Not in compliance with the inventory
PICCS	:	Not in compliance with the inventory
IECSC	:	On the inventory, or in compliance with the inventory
TSCA	:	This product is regulated under the United States Food and Drug Act (FDA).

Inventories

AIIC (Australia), DSL (Canada), IECSC (China), REACH (European Union), ENCS (Japan), ISHL (Japan), KECI (Korea), NZIoC (New Zealand), PICCS (Philippines), TCSI (Taiwan), TECI (Thailand), TSCA (USA)

Other regulations

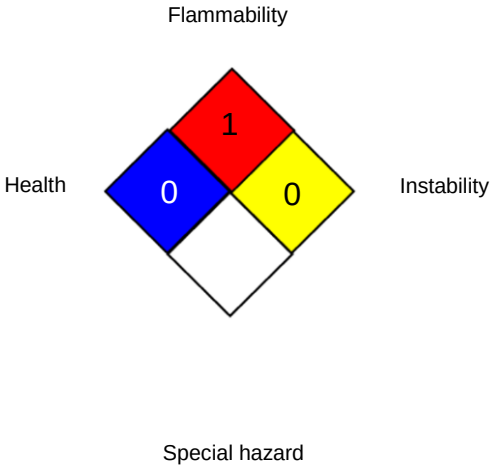
US. Drug Enforcement Administration (DEA) Listed	Not applicable

Precursor and Essential Chemicals (21 CFR 1310)

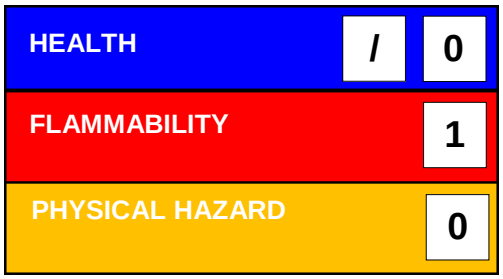
SECTION 16. OTHER INFORMATION

Further information

NFPA 704:



HMIS® IV:



HMIS® ratings are based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks. The "*" represents a chronic hazard, while the "/" represents the absence of a chronic hazard.

Full text of H-Statements

H302	Harmful if swallowed.
H318	Causes serious eye damage.
H335	May cause respiratory irritation.

Full text of other abbreviations

OSHA P0	:	USA. Table Z-1-A Limits for Air Contaminants (1989 vacated values)
OSHA Z-1	:	USA. Occupational Exposure Limits (OSHA) - Table Z-1 Limits for Air Contaminants
OSHA P0 / TWA	:	8-hour time weighted average
OSHA Z-1 / TWA	:	8-hour time weighted average

AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CERCLA - Comprehensive Environmental Response,

Compensation, and Liability Act; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DOT - Department of Transportation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; EHS - Extremely Hazardous Substance; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; HMIS - Hazardous Materials Identification System; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MSHA - Mine Safety and Health Administration; n.o.s. - Not Otherwise Specified; NFPA - National Fire Protection Association; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; RCRA - Resource Conservation and Recovery Act; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RQ - Reportable Quantity; SADT - Self-Accelerating Decomposition Temperature; SARA - Superfund Amendments and Reauthorization Act; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

This sds has been prepared by Ashland. (http://www.ashland.com)

Sources of key data used to compile the Material Safety Data Sheet	:	Ashland internal data including own and sponsored test reports European Union Law with content from the Official Journal of the European Union. European Chemicals Agency; the EU authority implementing the EU's chemicals legislation for companies. The German Water Hazard Classes. ReachCentrum; a series of support services to help comply with REACH regulations. The European Commission; proposing legislation, administering and implementing EU policies, and enforcing EU law. The UNECE administers regional agreements implementing

harmonised classification for labelling (GHS) and transport.
Cefic, the European Chemical Industry Council.
ESIS European Chemical Substances Information System

Revision Date : 10/25/2024

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.
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